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Executive Summary

This report presents the optimisation and validation of a lightweight bracket subject to three load cases. A topology optimisation study was conducted to minimise mass while maintaining structural performance, followed by design realisation for additive manufacturing using Powder Bed Laser Fusion (PBLF) and detailed finite element verification.

The final design achieves a mass reduction of approximately 49% ($0.94\,\mathrm{kg}$ to $0.48\,\mathrm{kg}$) compared to the baseline. Structural verification shows that all displacement requirements are satisfied, with a maximum bearing centre displacement below $1\,\mathrm{mm}$ in all load cases. The minimum factor of safety against yield is marginally above the required value of 1.2, governed by Load Case 2. The isotropy ratio is within acceptable limits, indicating reasonably consistent stiffness across loading directions.

It is recommended that the optimised design be accepted, as it meets all specified performance criteria while delivering a significant reduction in mass. The selected manufacturing route supports the production of the complex geometry while maintaining accessibility for inspection. However, the margin on strength is limited, and Load Case 2 remains critical. The design is therefore sensitive to modelling assumptions, particularly in the representation of the bearing interface, and further validation is advised.

The main risks associated with the design are the increased stress levels resulting from mass reduction, sensitivity to boundary condition assumptions, and the manufacturability of fine features inherent to topology optimisation. These risks can be mitigated through physical testing, refinement of the bearing interface modelling, and appropriate control of the additive manufacturing process.

Nomenclature

Quantity	Unit	Description
R_{iso}	—	Isotropy Ratio
E	GPa	Elastic Modulus
ν	—	Poisson's Ratio
σ	MPa	Stress
σ_y	MPa	Yield Stress
UTS	MPa	Ultimate Tensile Stress
c_i	$\mathrm{mm\,N^{-1}}$	Compliance for load case i
F	N	Applied Load
s	mm	Displacement
FoS	—	Factor of Safety
m	kg	Mass
V_f	—	Volume Fraction
p	—	Penalty Factor (SIMP)
$l_{\min}$	mm	Minimum Member Size

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1. Requirements & Assumptions

Summary of governing requirements, load cases, constraints, and key modelling assumptions.

The part to be optimised is a landing gear door bearing bracket on the Airbus A380. The initial mass of this component was $0.94403\,\mathrm{kg}$. The part is subject to three **LCs**, which are detailed in Figure 1. The material is 15-5 PH Stainless Steel, with material properties described in Table 1.

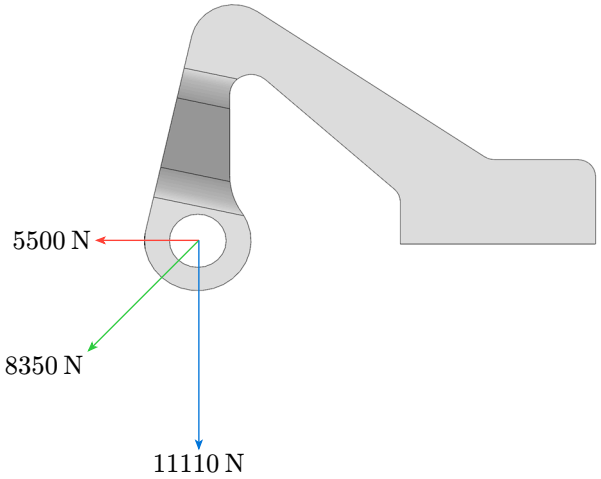


*Figure 1: Load Case Definitions - **LC1** is red, **LC2** is green, and **LC3** is blue. The arrows show the direction and magnitude of the applied loads in each case.*

TABLE 1
MATERIAL PROPERTIES FOR 15-5 PH STAINLESS STEEL IN THE H-1025 CONDITION [1]

Property	Symbol	Value	Notes
Material	—	15-5 PH Stainless Steel	AMS5862
Condition	—	H-1025	Heat treated condition
Elastic Modulus	E	$196\,\mathrm{GPa}$	—
Poisson's Ratio	ν	0.27	—
Yield Stress	σ_{yield}	$1000\,\mathrm{MPa}$	—
Ultimate Tensile Stress	UTS	$1069\,\mathrm{MPa}$	—
Density	ρ	$7810\,\mathrm{kg\,m^{-3}}$	—

For the purpose of analysis, the material is assumed to be isotropic and linear-elastic, the loads are assumed to be static, and the bolt holes are assumed to be rigidly fixed. The requirements for the optimised design are shown in Table 2.

TABLE 2
REQUIREMENTS FOR OPTIMISED DESIGN

Value	Limit	Comment
Minimum Factor of Safety	1.2	Against Yield Stress
Maximum Displacement At Bearing Centre	$1\,\mathrm{mm}$	$0.55\,\mathrm{mm}$ for high stiffness
Isotropy Ratio	2.45	2.2 desirable, 2 for high stiffness
Envelope	Must fit within original part envelope	

2. Baseline Assessment

The baseline analysis was conducted in Ansys Mechanical. For the purpose of the analysis, the bolts were modelled using fixed support boundary conditions at the boltholes to model a connection to a rigid plate. Loads were applied to the Ø16 mm bearing seat using a bearing load feature, with loads defined in components along the x and y axes. The bottom face of the bracket was unconstrained in order to maximise freedom of the optimiser.

A global element size of 2mm was used, with edge sizings applied on the boltholes and the curved edges of the bearing interface region to ensure a minimum of 16 elements. Face meshing was applied to the surfaces that do not contain holes. The resulting mesh had 71711 tetrahedral elements, with a minimum element quality of 0.126. A remote point was used to place a probe to monitor the bearing centre displacement. Initial analysis results are detailed in Table 3.

TABLE 3
BASELINE ANALYSIS RESULTS

Load Case	Max Stress (MPa)	Bearing Centre Displacement (mm)	Compliance (N mm)
1	374.9	0.489	2521
2	547.5	0.337	2280
3	522.0	0.539	4791

The highest stress occurs in **LC2**, while the highest displacement occurs in **LC3**. The stresses are significantly below the yield stress, and the displacements are small compared to the maximum allowable deformation.

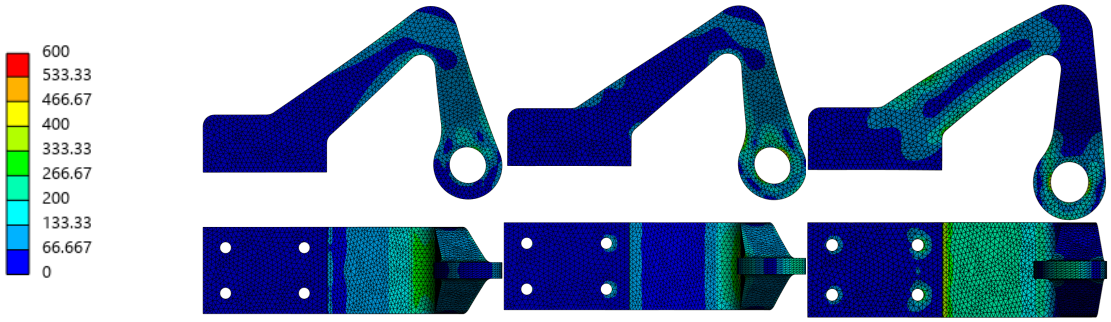


Figure 2: von Mises stress distribution for Load Cases 1 (left), 2 (centre), and 3 (right). All images are on the same scale, shown left (in MPa).

Figure 2 shows the von Mises equivalent stress and deformation distributions for each load case. It can be seen that the highest stresses are in the inner radius at the top of the arm, as well as the bearing seat. Uniquely, in **LC3**, the stress is highest at the root of the arm. The large block at the root of the arm shows minimal stress, suggesting the majority of the material in this region could be removed without significantly impacting the stiffness of the part. In addition, the lower section of the arm shows a low stress region through the centreline which could be reduced to form an I-beam-like cross-section or lightening holes could be introduced.

Figure 3 shows a that the region of highest stress in the Load Case 1 is located on the contact face with the bearing.

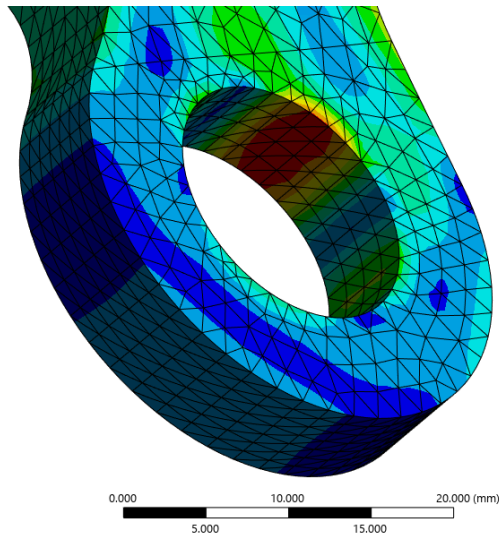


Figure 3: Detail showing stress concentration in bearing seat

3. Topology Optimisation

The three parameters selected for the exploratory optimisation were the volume fraction, the penalty factor, and the minimum member size. The volume fraction has the largest influence on the results, as it directly controls the allowable material removal and therefore governs the mass-stiffness trade-off. Lower volume fractions lead to increased compliance as load paths become more slender and less redundant. The penalty factor controls the penalisation of intermediate density elements within the SIMP formulation, influencing the discreteness of the solution. Higher values promote well-defined structural members, while lower values allow more diffuse material distributions that may reveal alternative load paths. The minimum member size was varied to investigate its influence on manufacturability and structural form. Larger values enforce thicker, more continuous features that are more suitable for conventional machining, while smaller values allow more complex, organic geometries typical of additive manufacturing. A complexity index was considered as an additional parameter, however, this is only available when using the level-set solver in Ansys and was therefore not included in this study. Its inclusion would primarily influence the geometric simplicity of the resulting design rather than the fundamental load path.

For **LC1**, the bolt-holes furthest from the arm were allowed to be removed to allow maximum freedom for the solver. The initial simulations were run with a target volume fraction of 0.5, a penalty factor of 4, and a minimum member size of $0.5\,\mathrm{mm}$. Table 4 shows the run matrix for each load case and varied parameter.

TABLE 4
RUN MATRIX FOR INITIAL EXPLORATORY RUNS

Run ID	Load Case	Varied Parameter	Value	Compliance (N mm)	Mass (kg)
1A	1	Volume Fraction	0.40	3004.33	0.38941
1B	1	Penalty Factor	3	2652.35	0.38986
1C	1	Min. Member Size	1.5	2652.35	0.38986
2A	2	Volume Fraction	0.3	4134.14	0.28484
2B	2	Penalty Factor	2	3305.42	0.28406
2C	2	Min. Member Size	6	4052.28	0.28656
3A	3	Volume Fraction	0.37	5803.56	0.35738
3B	3	Penalty Factor	2	4989.53	0.36088
3C	3	Min. Member Size	3	5028.37	0.35862

In Run 1C, increasing the minimum member size had no effect until it exceeded $10\,\mathrm{mm}$, at which point the compliance and mass both increased significantly to $3406.32\,\mathrm{N\,mm}$ and 0.39915 kg respectively, as the structure resembled an I-beam instead of a space frame. The solution from Run 1C is therefore identical to 1B.

3.1. Volume Fraction Variation

Figure 4 shows the variation of compliance with mass for runs 1A, 2A, and 3A, where the volume fraction was varied. The compliances are normalised by the applied force in each load case to allow direct comparison. All load cases exhibit increasing compliance with decreasing mass, reflecting the expected trade-off between stiffness and material usage.

The selected points from the run matrix are also shown, illustrating the chosen balance between mass reduction and structural performance. These points lie within the transition region of the curves, where significant mass savings can be achieved without a disproportionate increase in compliance. The points in the run matrix were selected to be as close to the “High Stiffness” line - in **LC1**, the point closest to the line was selected, but in **LC2** and **LC3**, the selected points were approximately halfway between the two points that straddle the line.

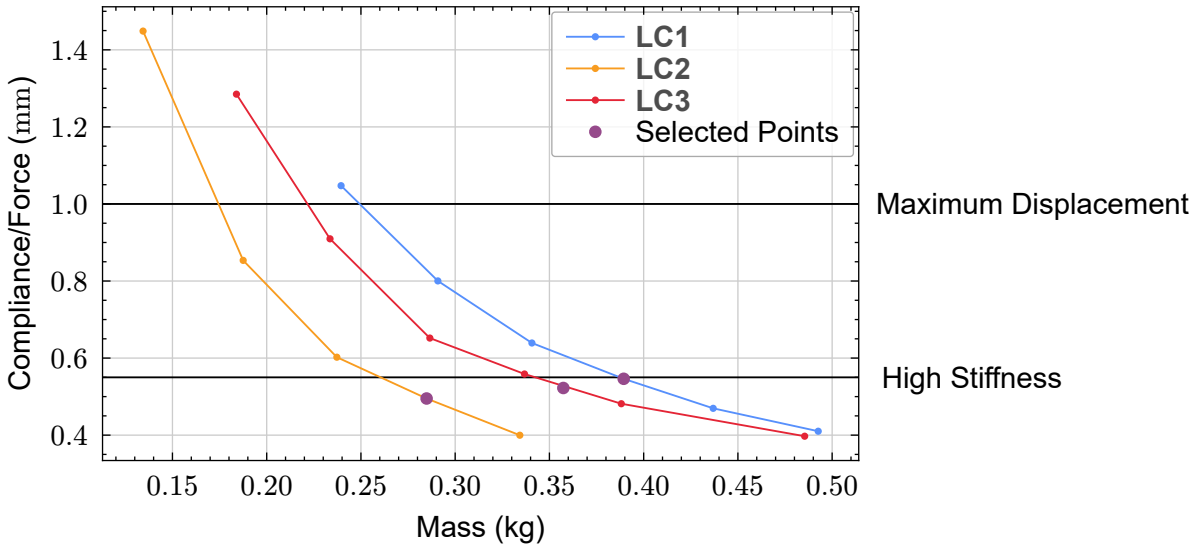


*Figure 4: Mass vs Compliance for Volume Fraction Variation. Selected Points show final choice for each **LC***

3.2. Penalty Factor and Minimum Member Size Variation

In Table 4, it is shown that decreasing the penalty factor from 3 to 2 in **LC2** and **LC3** resulted in a significant reduction in compliance in exchange for a small increase in mass.

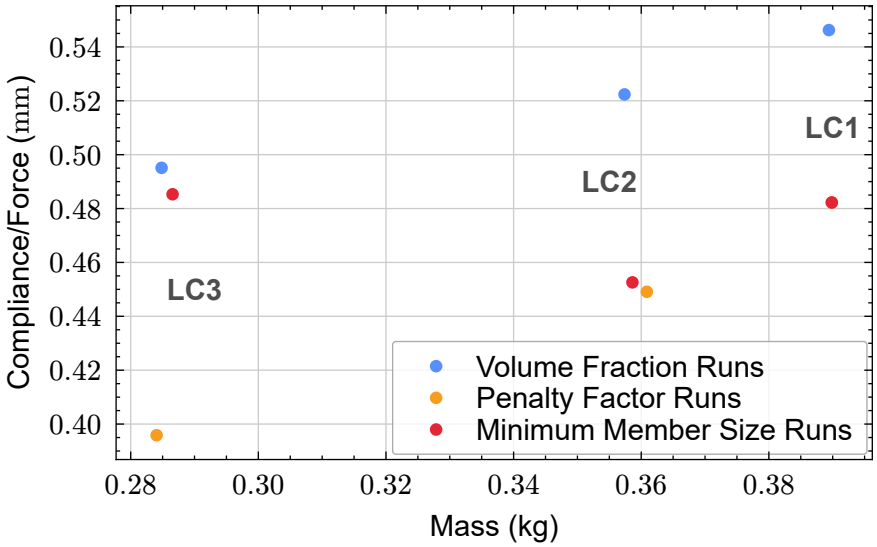


*Figure 5: Mass vs Compliance for All Parameter Variations. Points are coloured by parameter varied, and grouped by **LC**.*

Figure 5 shows that, for a given mass, the runs with lower penalty factors generally achieve lower compliance than the corresponding volume fraction-only runs, indicating improved structural efficiency. This suggests that further mass reduction may be possible for a given stiffness level through appropriate tuning of the penalty factor. However, lower penalty values promote intermediate density elements, resulting in more diffuse material distributions and increased geometric complexity, which may reduce manufacturability. The same trend is not present in all cases when varying the minimum member size, suggesting that the bulkier results from larger minimum member sizes are not always equally efficient as a more organic design with a smaller minimum member size.

3.3. Final Outputs

Figure 6 shows the optimised geometries for each load case,



Figure 6: Outputs from Run 1C (top), Run 2C (middle), and Run 3C (bottom), showing side (left), front (middle), and top (right) views.

4. Design Realisation

The final design is shown in Figure 7 inside the design envelope. It was mostly inspired by the TO results from Run 3C, with the vertical cut through the arm of the bracket being inspired by Run 2C. The results of Run 1C suggested that the full width of the body should be retained, and contradicted the idea of introducing a lightening hole above the bearing seat suggested by the results of Run 3C.

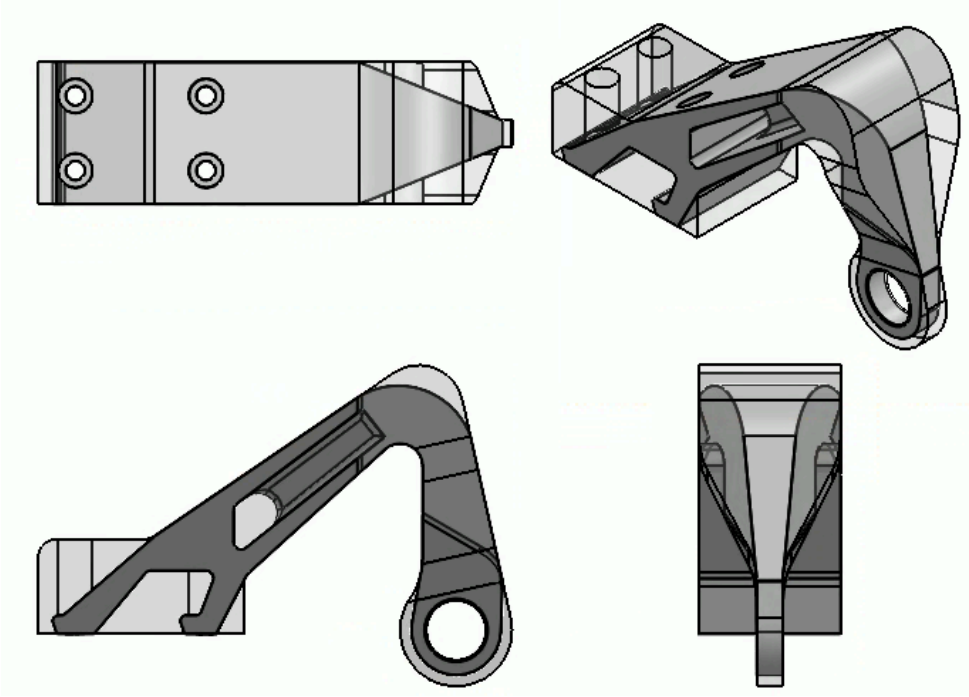


Figure 7: Final design concept inside the design envelope.

The design is intended to be 3D-Printed via Powder Bed Laser Fusion (**PBLF**). The face in contact with the bed would be the flat face on the front of the arm. Figure 8 shows the slicer output from Ultimaker Cura for the final design, with support overhang angle set to $50\,°$. Support material is shown in teal. It should be noted that the slicer used for the generation of these images is set to presets relevant to Fused Filament Fabrication (**FFF**) of PLA, therefore layer heights and top surfaces will appear differently in **PBLF**. Approximately 15% of the print time is associated with support generation. The chosen orientation avoids large horizontal cross-sections, reducing the risk of residual stress accumulation and thermal distortion, and minimises the horizontal space used so multiple parts can be printed simultaneously. In addition, the dominant load case is applied at approximately $40\,°$ from the build direction, and the largest load is perpendicular to the build direction, which should help to mitigate the anisotropy of the part.

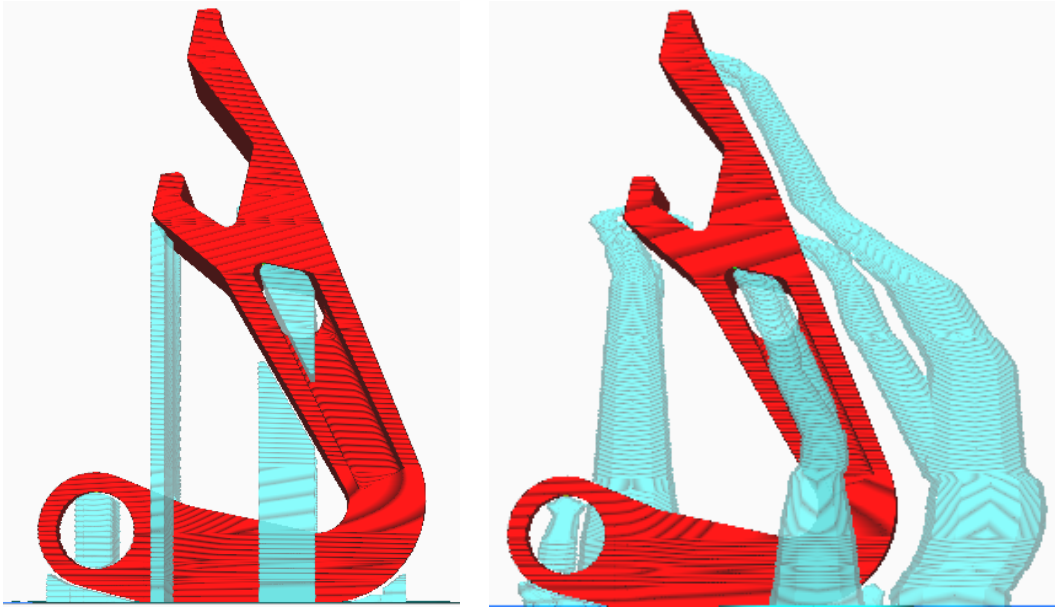


Figure 8: Slicer output for final design, showing support material in teal. Normal (left) and Tree (right) support generation strategies are shown.

Support material is required in the lightening holes in the I-beam web, the leg closer to the arm, in the bearing seat, and for the boltholes themselves. Both the Normal and Tree support strategies result in print times of approximately eleven hours using an **FFF** printer on some relatively fine settings.

Access for non-destructive testing (NDT) was considered, and intersecting holes were avoided. All critical regions of the part are accessible for visual and dye penetrant inspection. The component is also sufficiently open and not excessively bulky, making radiographic or Computed Tomography (**CT**) inspection feasible. In addition, 15-5 PH stainless steel is suitable for magnetic particle inspection after heat treatment, allowing surface and near-surface defects to be detected in critical regions such as the bearing seat and boltholes.

5. Design Verification

The following subsections show mesh convergence studies and final results for each load case. Each simulation used fixed supports in the boltholes, and a bearing load in the mount face.

5.1. Load Case 1

Table 5 and Figure 9 show the results of a mesh convergence study for Load Case 1. Major refinement areas in this case were the underside of the apex of the arm, and the web of the I-beam-like section. The results show that the maximum von Mises stress and bearing centre displacement converge to within ±5% of the final value at approximately 10,000 elements, but the number of elements was increased to approximately 70,000 to ensure a smooth stress distribution in the refined areas. The final mesh used had 72,231 elements and 126,827 nodes.

TABLE 5
MESH CONVERGENCE STUDY FOR FINAL DESIGN LOAD CASE 1

Nodes	Elements	Maximum vM Stress	Bearing Deflection
8777	4527	538.61	0.8304
10518	5474	635.84	0.83898
12148	6368	625.46	0.8398
16900	8939	650.45	0.84244
28393	15844	641.78	0.84398
28917	16132	648.48	0.84412
30944	17277	677.8	0.84438
34605	19387	652.97	0.84478
35210	19725	662.07	0.84476
35715	20060	635.25	0.84433
37085	20894	636.29	0.84492

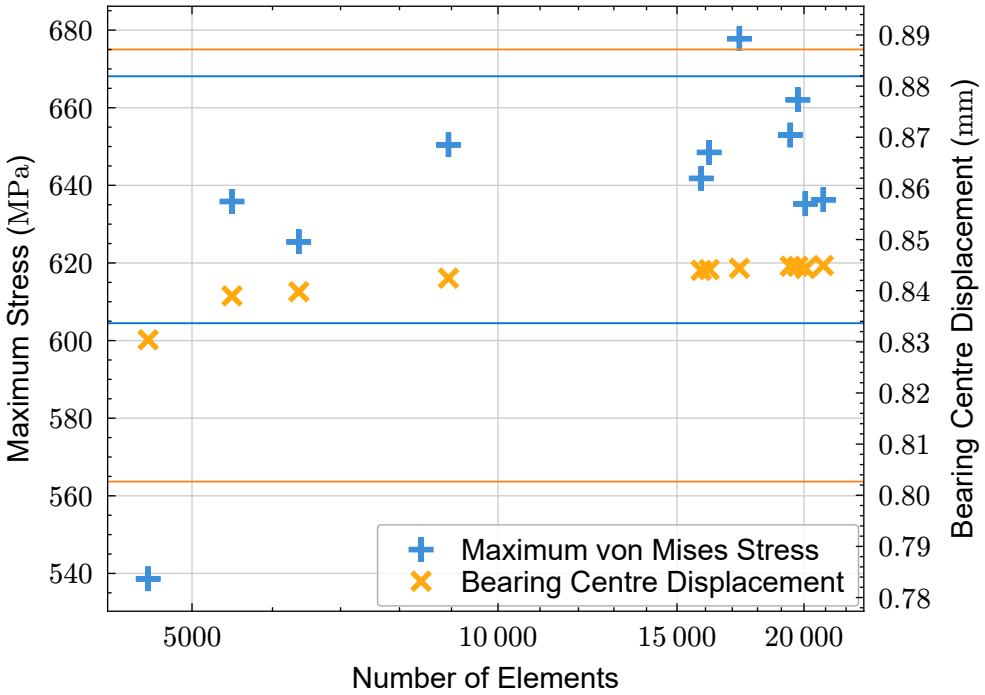


Figure 9: Mesh Convergence Study for Final Design in Load Case 1. Blue and orange lines show ±5% bounds for maximum stress and bearing centre displacement respectively.

The maximum von Mises equivalent stress was $636.29\,\mathrm{MPa}$, giving a safety factor of 1.57 against yield, higher than the minimum of 1.2. The bearing centre displacement was 0.84492 mm, which is below the maximum of $1\,\mathrm{mm}$.

5.2. Load Case 2

Table 6 and Figure 10 show the results of a mesh convergence study for Load Case 2.

TABLE 6
MESH CONVERGENCE STUDY FOR FINAL DESIGN LOAD CASE 2

Nodes	Elements	Maximum vM Stress	Bearing Deflection
8777	4527	835.6	0.69063
9620	4994	869.62	0.69115
21957	12248	869.4	0.70172
26625	14829	879.78	0.70419
31708	17767	879.91	0.70481
51538	29359	887.17	0.70546
56259	31891	886.77	0.70602

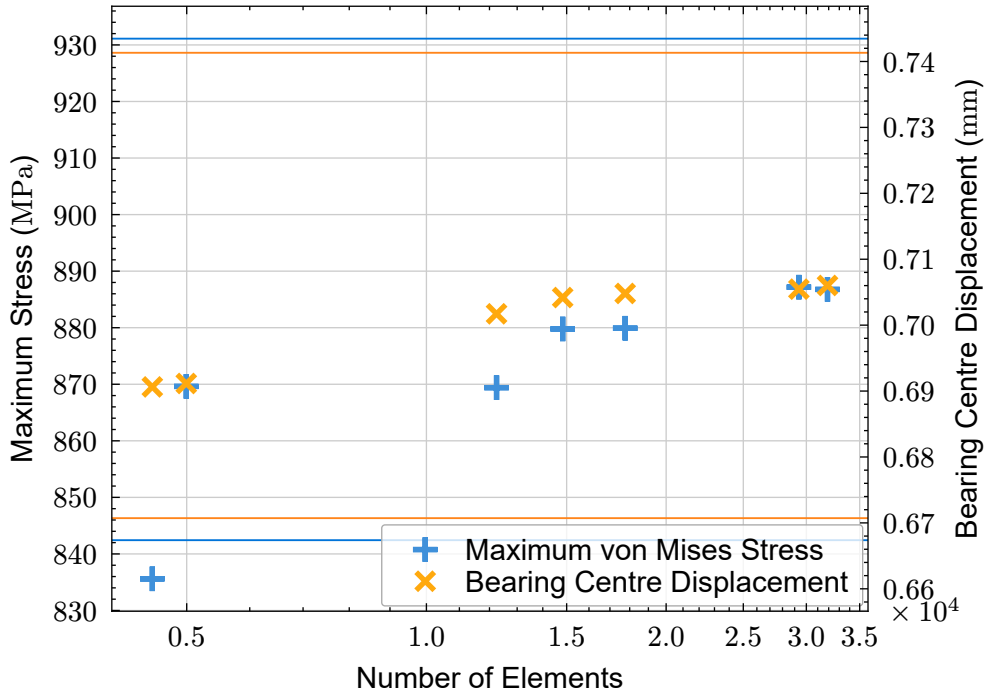


Figure 10: Mesh Convergence Study for Final Design in Load Case 2. Blue and orange lines show ±5% bounds for maximum stress and bearing centre displacement respectively.

The maximum von Mises equivalent stress was $886.77\,\mathrm{MPa}$, giving a safety factor of 1.13 against the yield stress, lower than the minimum of 1.2. The bearing centre displacement was $0.70602\,\mathrm{mm}$, which is below the maximum of $1\,\mathrm{mm}$.

The point of maximum stress in this case occurred at the bearing seat, as shown in Figure 11. This localised peak was considered overly conservative, as the stiff spherical bearing would in practice limit deformation of the seat and redistribute the applied load over the interface.

To better represent this behaviour, a rigid remote displacement condition was applied to the bearing seat, enforcing compatibility across the interface. Minimal rotational constraints were introduced to fully define the boundary condition, but do not significantly influence the load transfer. Table 7 and Figure 12 show the results of a mesh convergence study using this revised boundary condition. The results from this revised model were used for final compliance assessment.

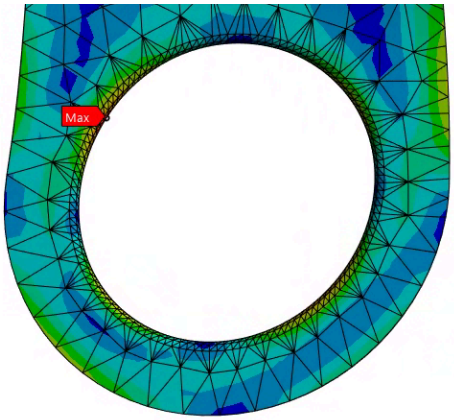


Figure 11: Detail showing stress concentration in bearing seat in Load Case 2

TABLE 7
MESH CONVERGENCE STUDY FOR FINAL DESIGN LOAD CASE 2 WITH RIGID BEARING SEAT

Nodes	Elements	Maximum vM Stress	Bearing Deflection
8777	4527	741.32	0.62693
14741	7902	710.98	0.63678
17975	9806	719.04	0.63566
27479	15193	725.17	0.63835
48250	26998	729.52	0.64098
49669	27726	732.9	0.64157

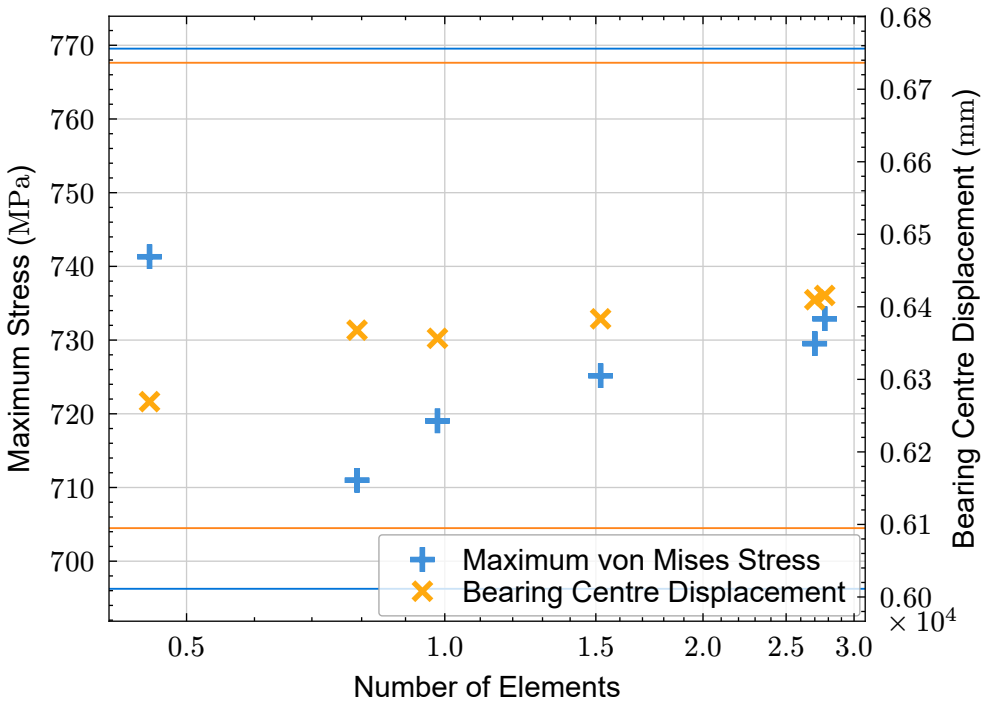


Figure 12: Mesh Convergence Study for Final Design in Load Case 2 with rigid bearing seat. Blue and orange lines show ±5% bounds for maximum stress and bearing centre displacement respectively.

The maximum von Mises equivalent stress was $732.9\,\mathrm{MPa}$, giving a safety factor of 1.36 against the yield stress. The bearing centre displacement, was $0.64157\,\mathrm{mm}$, which is below the maximum of $1\,\mathrm{mm}$.

5.3. Load Case 3

Table 8 and Figure 13 show the results of a mesh convergence study for **LC3**. The added remote displacement constraint was carried over from **LC2**.

TABLE 8
MESH CONVERGENCE STUDY FOR FINAL DESIGN LOAD CASE 3

Nodes	Elements	Maximum vM Stress	Bearing Deflection
8777	4527	587.58	0.72915
10122	5240	693.61	0.73084
13752	7249	690.36	0.73227
24828	13513	694.31	0.72248
38168	21447	699.75	0.72479
42541	23865	698.97	0.72533

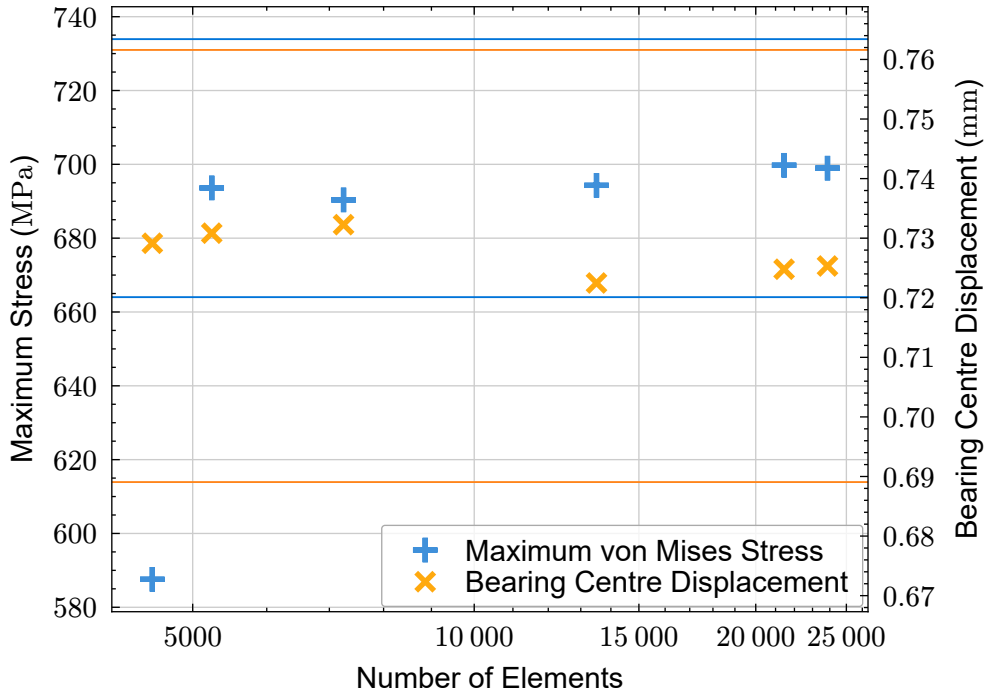


Figure 13: Mesh Convergence Study for Final Design in Load Case 3. Blue and orange lines show ±5% bounds for maximum stress and bearing centre displacement respectively.

The maximum von Mises equivalent stress was $698.97\,\mathrm{MPa}$, giving a safety factor of 1.43 against the yield stress. The bearing centre displacement, was $0.72533\,\mathrm{mm}$, which is below the maximum of $1\,\mathrm{mm}$.

5.4. Isotropy Ratio

The Isotropy Ratio is calculated in Equation 1, where c_i is the ratio of bearing centre displacement to applied load in each load case.

$$R_{\text{iso}} = \frac{\max(c_i)}{\min(c_i)} \tag{1}$$

The isotropy ratio for the final part is given in Equation 2.

$$R_{\text{iso}} = \frac{\max(c_i)}{\min(c_i)} = \frac{153.6 \times 10^{-6}}{65.29 \times 10^{-6}} = 2.353 \tag{2}$$

The maximum value of c_i is in **LC1**, and the minimum value is in **LC2**. The isotropy ratio of 2.353 is below the requirement of 2.45, but above the desirable value of 2.2, suggesting the part is reasonably isotropic, but there is some scope for improvement in this area.

5.5. Final Values

The important values for the final design are summarised in Table 9. The dominant load case is **LC2**, which has the lowest stress safety factor. While a high-stiffness solution was sought from the outset, the final geometry does not fully achieve this objective. However, all specified requirements on displacement, safety factor, and isotropy ratio are satisfied by the final design.

TABLE 9
FINAL DESIGN RESULTS

Quantity	Value
Mass	$0.48137\,\mathrm{kg}$
% Reduction	49.01%
R_{iso}	$2.2 < 2.353 < 2.45$
Minimum Safety Factor	$1.36 > 1.2$
Maximum Displacement	$0.84492\,\mathrm{mm}$
Dominant **LC**	2

The results of analyses compared to the baseline are shown in Table 10.

TABLE 10
COMPARISON OF ANALYSES WITH BASELINE

	LC1		**LC2**		**LC3**	
	σ (MPa)	s (mm)	σ (MPa)	s (mm)	σ (MPa)	s (mm)
Original	374.9	0.489	547.5	0.337	522.0	0.539
Optimised	636.3	0.844	732.9	0.642	699.0	0.725
% Difference	69.7%	72.6%	33.9%	90.5%	33.9%	34.5%

This comparison shows that while the optimised design achieves a significant mass reduction, at the cost of increased von Mises equivalent stresses and bearing centre displacements in every load case. **LC2** governs the strength assessment, giving the minimum safety factor. **LC1** governs the maximum displacement response per unit load and therefore the isotropy ratio. **LC3** produces the largest absolute applied load, but its response remains less critical than **LC2** in terms of strength.

6. Durability & Robustness

Assessment of fatigue, damage tolerance, or other non-ideal behaviour, including implications for design robustness.

The S-N curves shown in Figure 14 were digitised from data from Boeing [2] for 15-5 PH stainless steel in the H-900 and H-1050 conditions, both unfinished and shot peened.

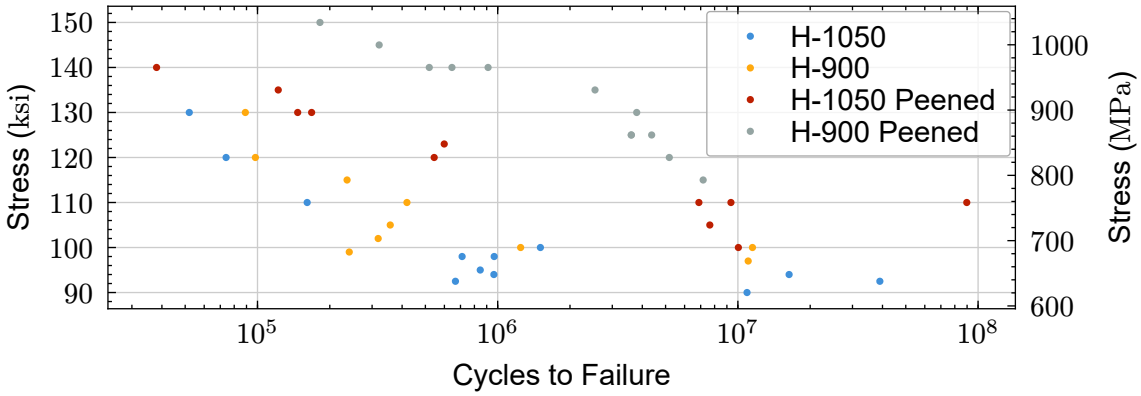


Figure 14: S-N Curves for 15-5 PH Stainless Steel in H-900 and H-1050 Conditions, both Unfinished and Shot Peened

The maximum stress experienced by the part is $732\,\mathrm{MPa}$ or $106.2\,\mathrm{ksi}$, obtaining a lifetime of approximately $1 \times 10^5\,\mathrm{cycles}$ for unfinished H-1050 and H-900, and approximately $9 \times 10^6\,\mathrm{cycles}$ for a shot peened finish. Assuming each aircraft conducts two flights per day, and the doors are actuated twice per flight (closing on take-off, opening on approach), this fatigue life corresponds to approximately 68 years of service for an unfinished part, and 6160 years for a shot peened part. The part is therefore unlikely to experience fatigue failure during its service life, but a shot peened finish would provide additional safety margin and durability.

It should be noted that the fatigue data used is for machined specimens. Local stress concentrations, residual stresses, and general weakening from laser fusion may reduce the fatigue life without further post-processing such or shot peening or heat treatment.

7. Risks & Recommendations

A risk register for the optimised design is shown in Table 11.

TABLE 11
RISK REGISTER FOR OPTIMISED DESIGN

Risk	Likelihood	Impact	Mitigation
Residual Stresses from AM	Medium	High	Post-process via HIP-ing or shot peening
Crack Initiation	Medium	High	Inspect high-stress areas regularly
Porosity Defects from AM	Medium	High	Conduct **CT** scans before parts enter service
Dimensional Inaccuracies	Low	Medium	Calibrate AM machine and conduct quality checks on first parts

Prior to production release, the optimised design should undergo a validation programme to verify fatigue performance, dimensional accuracy, and manufacturing quality. Additively manufactured material coupons should first be tested to establish baseline mechanical and fatigue properties for the **PBLF** process. Radiographic or **CT** inspection should then be conducted on the manufactured component to identify internal defects such as porosity or lack-of-fusion, particularly within thicker load-bearing regions. Dimensional inspection of critical interfaces, including the bearing seat and bolt holes, should be performed following post-processing and machining to ensure compliance with tolerance requirements. Finally, a component-level fatigue test should be conducted to validate the predicted fatigue life, with particular focus on the web of the arm where elevated stresses were identified in the simulation.

Bibliography

[1] C. Triantafyllou, ‘Project WINGTIP: Weight-Integrated Next-Generation Topology Improvement Programme’. 2026.

[2] M. Raefsky, ‘Fatigue Properties of 17-4 PH and 15-5 PH Steel in the H-900 and H-1050 Condition’. [Online]. Available: https://apps.dtic.mil/sti/citations/AD0691794